power

File: power.kicad_sch

walter

File: walter.kicad_sch

usb_serial

File: usb_serial.kicad_sch

display

File: display.kicad_sch

peripherals

File: peripherals.kicad_sch

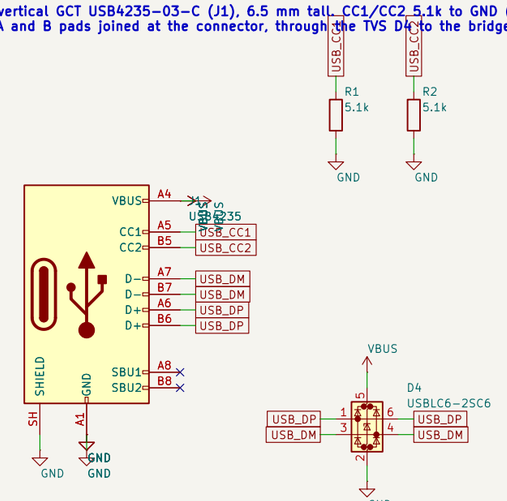
awong-dev/pager

Sheet: /
File: pager-carrier.kicad_sch

Title: pager-carrier

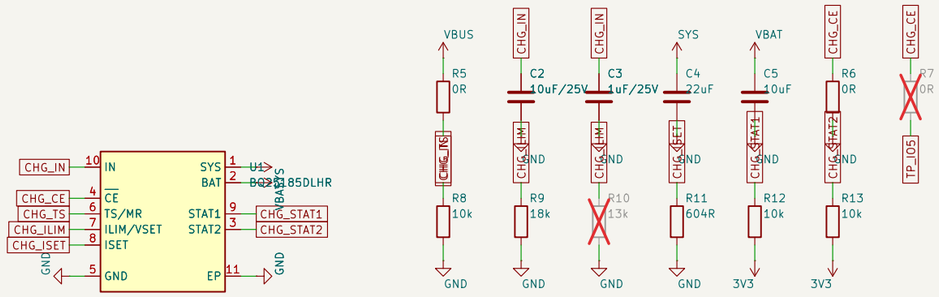
Size: A4	Date: 2026-10-17	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/6

USB-C: vertical GCT USB4235-03-C (J1), 6.5 mm tall. CC1/CC2 5.1k to GND (UFP).
D+/D- A and B pads joined at the connector, through the TVS D4 to the bridge U5.



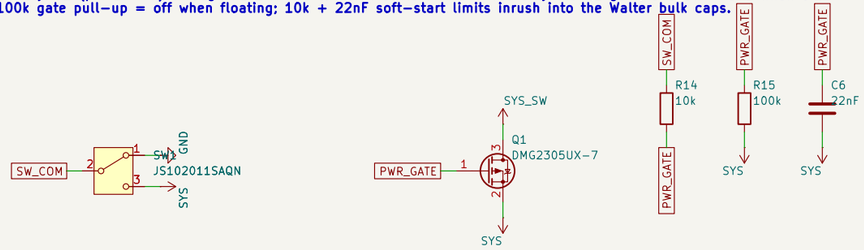
D4: flow-through TVS. Connector pads into pins 1/3, out of pins 6/4 toward U5.

Charger BQ25185DLHR: ISET 604R = 497 mA fast charge. ILIM/VSET 18k = 500 mA input limit, 4.20 V (13k DNP = 1.1 A).
TS unused: 10k to GND. CE is active-low: 0R to GND. STAT1/2 open-drain, 10k to 3V3.

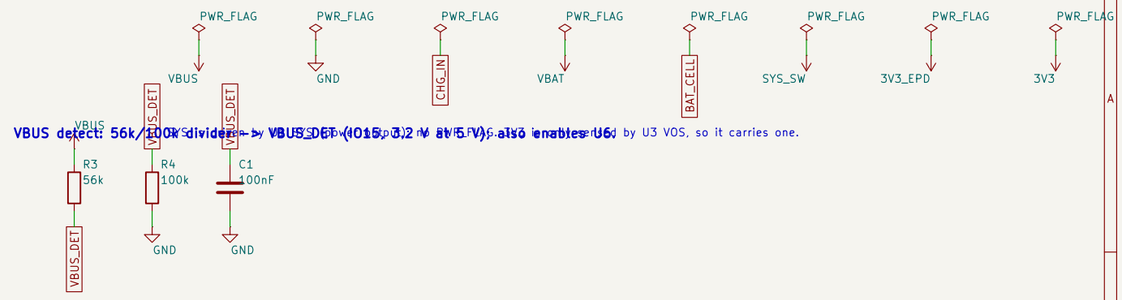




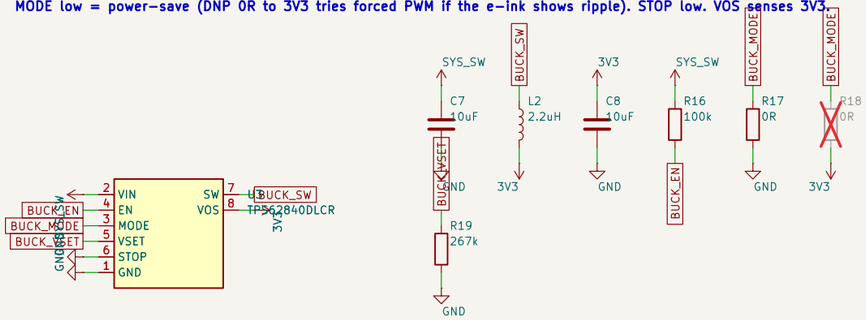
Battery: J2 (pin 1 = +) through F1 2 A PTC to VBAT. Switch: SW1 drives only the Q1 gate; SYS_SW = SYS when on.
100k gate pull-up = off when floating; 10k + 22nF soft-start limits inrush into the Walter bulk caps.

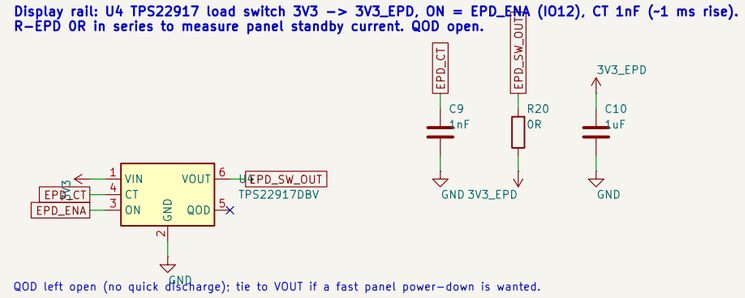


SW1: common (pin 2) -> SW_COM. Pin 1 -> GND = ON. Pin 3 -> SYS = OFF.

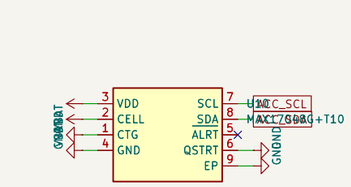


VBUS detect: 56k/100k divider -> VBUS_DET (~3.2 V at 5 V), also enables U6.





QOD left open (no quick discharge); tie to VOUT if a fast panel power-down is wanted.

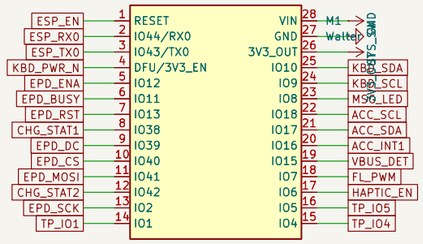


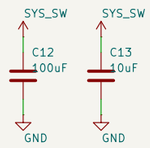
awong-dev/pager
Sheet: /power/
File: power.kicad_sch

Title: USB-C input, BQ25185 charger, battery switch, 3.3 V rails

Size: A3	Date: 2026-10-17	Rev: A
KiCad E.D.A. 10.0.6		Id: 2/6

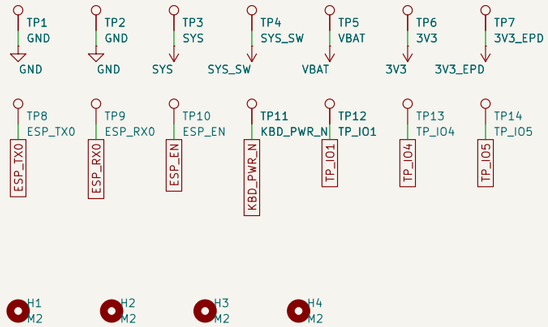
Walter 2x14 header. Pin names as datasheet rev 1.1 Table 1. The module is mounted header-side-up under the carrier; the footprint is the mirror of the vendor top-mount pattern.





Bulk at Walter VIN (pin 28): 100uF 1210 + 10uF 0603, within 5 mm of the pin.

Test pads, 1.0 mm, top side.



IO0 (pin 4, KBD_PWR_N) is a strap, the keyboard-rail gate and the auto-reset boot target; see usb_serial.

Spare IO1/IO4/IO5 go to test pads only (TP_IO1 = ADC1_CH0). IO5 also has a DNP 0R to the charger CE.

awong-dev/pager

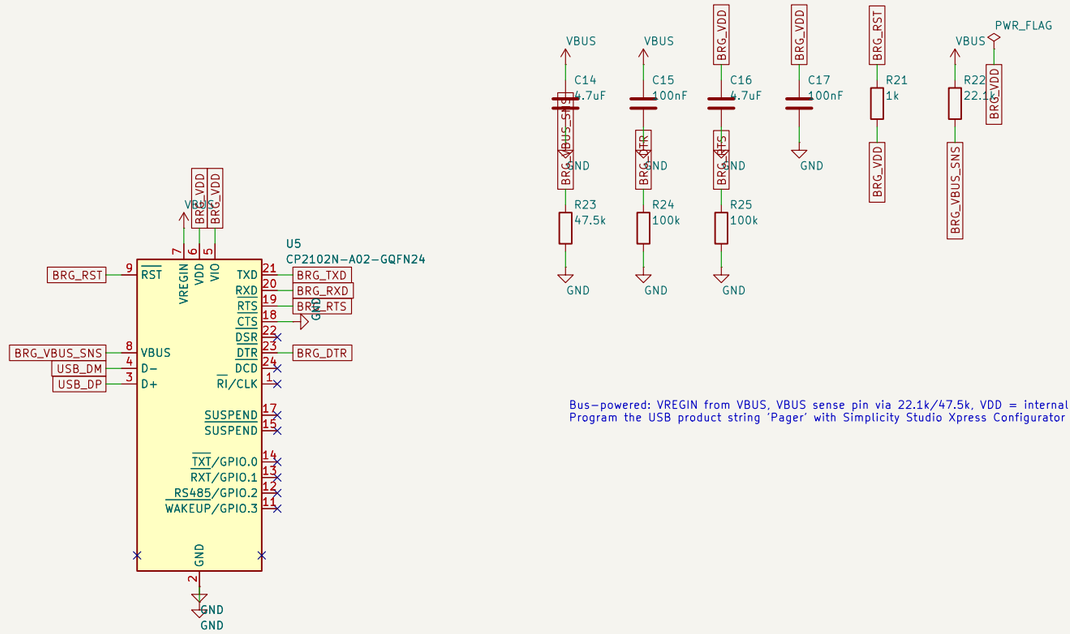
Sheet: /walter/

File: walter.kicad_sch

Title: Walter module header, bulk capacitance, test pads, mounting holes

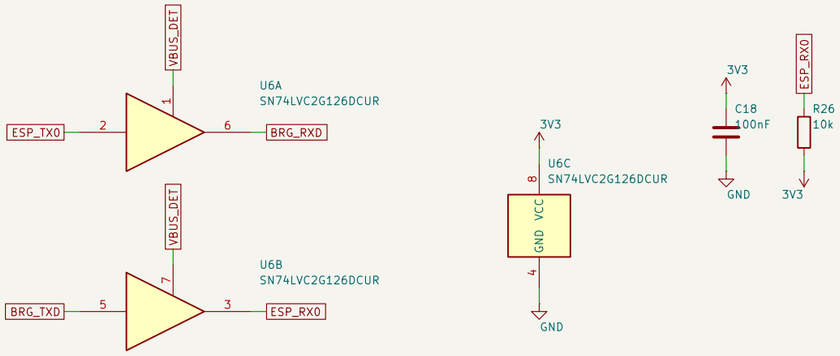
Size: A3	Date: 2026-10-17	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 3/6

USB-serial bridge: bus-powered CP2102N (unpowered without a cable), SN74LVC2G126 isolates UART0 when VBUS_DET is low, Espressif two-transistor auto-reset.

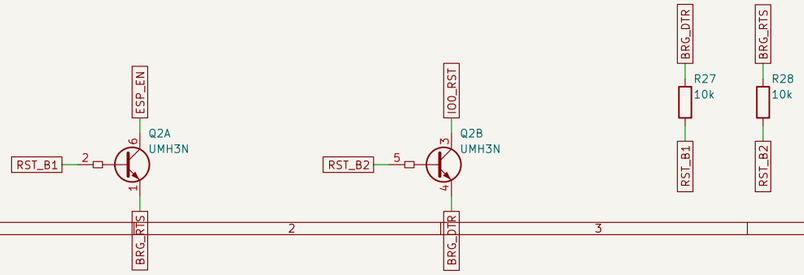


Bus-powered: VREGIN from VBUS, VBUS sense pin via 22.1k/47.5k, VDD = internal 3.3 V regulator output, VIO tied to VDD, 4.7uF + 100nF per power pin.
Program the USB product string 'Pager' with Simplicity Studio Xpress Configurator after assembly.

U6 SN74LVC2G126: both OE from VBUS_DET. A: ESP_TX0 -> BRG_RXD. B: BRG_TXD -> ESP_RX0. 10k keeps ESP_RX0 high with no cable.

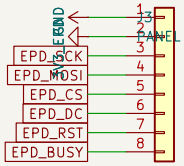


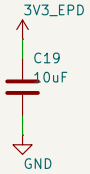
Auto-reset (Espressif): Q2 DTR->EN, Q3 RTS->IO0. JP1 (closed) in Q3's collector lets auto-boot be disabled; reset still works.



awong-dev/pager
Sheet: /usb_serial/
File: usb_serial.kicad_sch
Title: CP2102N USB-UART bridge, isolation buffer, auto-reset
Size: A3 | Date: 2026-10-17 | Rev: A
KiCad E.D.A. 10.0.6 | Id: 4/6

The SSD1680 ZIF and boost/charge-pump cluster live on the separate panel board (hardware/carrier/panel). The carrier carries only the 8-pin JST SH cable connector J3.

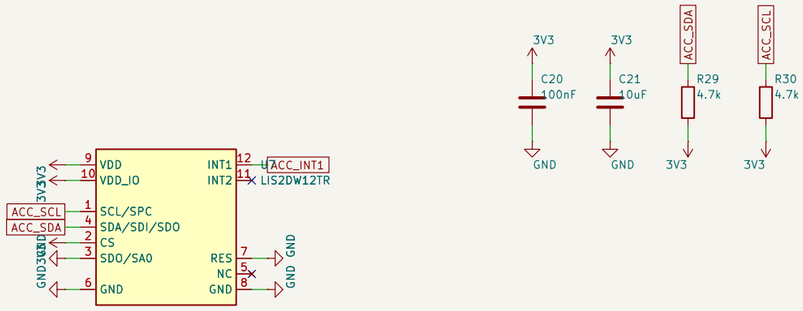




Pinout: 1 3V3_EPD, 2 GND, 3 EPD_SCK, 4 EPD_MOSI, 5 EPD_CS, 6 EPD_DC, 7 EPD_RST, 8 EPD_BUSY. The panel board carries the RST/CS pull-ups and the VCI decoupling.

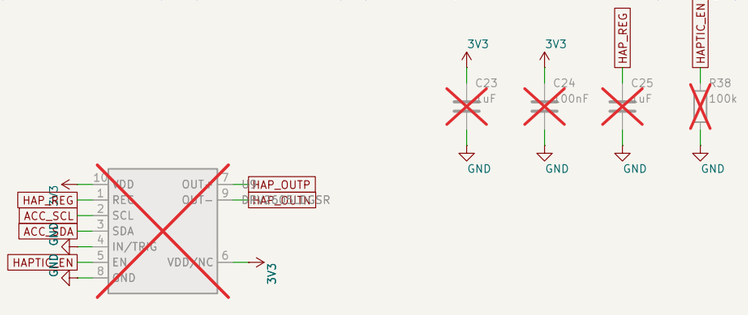
awong-dev/pager		
Sheet: /display/		
File: display.kicad_sch		
Title: Display cable to the panel board		
Size: A4	Date: 2026-10-17	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 5/6

U7 LIS2DW12 on I2C1 (ACC_*), SA0 low = 0x18, CS high = I2C mode, INT1 -> IO16 (RTC wake). Always on from 3V3.

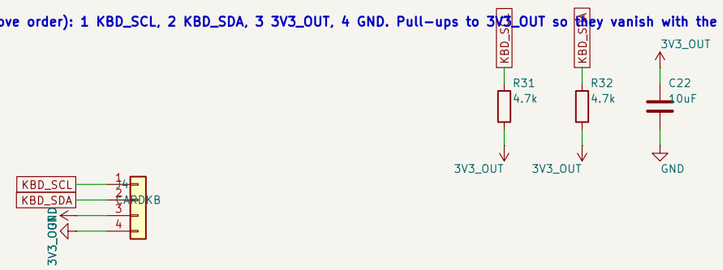


Place U7 away from J7 and the panel cable; X axis along the tin's long edge; mark orientation on silk.

Haptic (DNP): DRV2605L on I2C1 (0x5A), EN = HAPTIC_EN (IO6) with 100k pull-down, IN/TRIG to GND (I2C trigger), OUT+/- to J7.

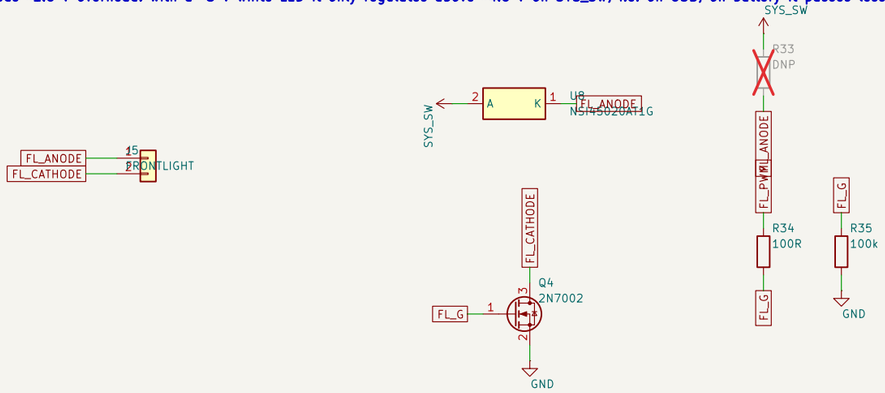


J4 CardKB (Grove order): 1 KBD_SCL, 2 KBD_SDA, 3 3V3_OUT, 4 GND. Pull-ups to 3V3_OUT so they vanish with the rail.

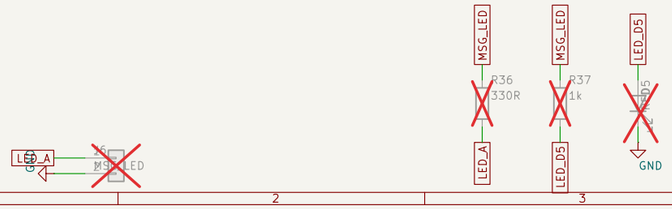




Front light: U8 20 mA CCR (anode pin 2 from SYS_SW, cathode pin 1 -> J5 pin 1); Q4 low-side switch on J5 pin 2, PWM from FL_PWM (IO7). R-FL DNP alt.
NSI45020 needs ~1.8 V overhead: with a ~3 V white LED it only regulates above ~4.8 V on SYS_SW, i.e. on USB; on battery it passes less than 20 mA.



Message LED (DNP): MSG_LED (IO8, RTC GPIO) -> 330R -> J6 pin 1; J6 pin 2 GND. D5 + 1k on-board bench LED on the same net (DNP).



awong-dev/pager

Sheet: /peripherals/
File: peripherals.kicad_sch

Title: Accelerometer, keyboard, front light, message LED, haptic

Size: A3	Date: 2026-10-17	Rev: A
KiCad E.D.A. 10.0.6		Id: 6/6